

Jonathan Ma

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LinkedIn | GitHub | Personal Website

Education

Master of Science in Engineering, Applied Mathematics and Statistics Aug 2025 – Dec 2026
Johns Hopkins University Baltimore, MD

- Coursework: Elements of Statistical Learning, Nonlinear Optimization, Probabilistic Machine Learning, Non-parametric Bayesian Statistics, AI and Statistical Methods in Clinical Research

Bachelor of Science in Finance, Bachelor of Arts in Economics, Minor in Statistics Aug 2021 – May 2025
Rutgers University Camden, NJ

- Coursework: Econometrics, Applied Data Mining and Machine Learning, Regression Analysis and Time Series, Applied Statistics, Statistical Models, Data Visualization, Textual Analysis in Finance, Machine Learning for Business Applications
- Honors: Summa Cum Laude, Honors in Economics, Aetheneum Honors Society

Research Experience

Affiliate Researcher, Sharon Eccles Steele Center for Translational Medicine Jun 2026 – Aug 2026
University of Utah Salt Lake City, UT

- Developed computational pipelines to detect and quantify hypertransmission (“barcoding”) patterns in optical coherence tomography (OCT) images of age-related macular degeneration (AMD), integrating Heidelberg Eye Explorer (HEYEX) segmentations to construct a weighted barcoding index from quantitative imaging features, with feature weights estimated using logistic regression to maximize progression-risk prediction.
- Designed and evaluated classical image analysis and deep learning approaches for automated hypertransmission localization, including transfer-learning CNNs (ResNet50), bounding-box detection strategies, and Grad-CAM explainability on over 24,000 retinal OCT images, achieving approximately 93% classification accuracy after layer-wise fine tuning.
- Performed robustness and sensitivity analyses through ROI perturbation, bootstrap validation, and segmentation reliability testing while collaborating with ophthalmologists, clinicians, and biostatisticians to develop clinically interpretable biomarkers, clinician-assisted annotation software, and HIPAA-compliant workflows for secure imaging analysis and progression-risk modeling.

Research Software Developer, Modern AutoCriteria May 2026 – Present
Johns Hopkins University Baltimore, MD

- Refactored the AutoCriteria clinical eligibility extraction platform into modular, maintainable components by replacing LangChain dependencies with the OpenAI SDK, implementing structured logging, token usage monitoring, adaptive prompting across multiple foundation models, and automatic retry workflows for API rate limiting.
- Developed terminology normalization and semantic preprocessing pipelines integrating ICD-10 and RxNorm vocabularies with regular-expression normalization, abbreviation expansion, and similarity-based matching to improve consistency of extracted eligibility criteria and support downstream meta-analysis and evidence synthesis.
- Designed and implemented an independent validation agent that compares extracted eligibility criteria against the original ClinicalTrials.gov XML records, automatically identifying omissions, inconsistencies, and hallucinated entities while improving extraction performance to an Exact Match score of 91.8% and an F1 score of 0.96.
- Expanded the platform with a dynamic terminology dictionary, multi-model support, automated model integration testing, structured export pipelines, and both command-line and Tkinter desktop interfaces to improve reproducibility, usability, and future clinical deployment.

Research Assistant, Dr. Tengfei Zhang (Finance)

Oct 2023 – Jul 2025

Rutgers University

- Developed an end-to-end financial text analytics pipeline by collecting and preprocessing over 7,500 earnings call and conference transcripts from Capital IQ using Python and R, producing a structured corpus for downstream sentiment and disclosure analysis.
- Developed a FinBERT-assisted sentiment analysis workflow incorporating structured prompt engineering and manual quality assurance to improve consistency and reliability of firm-level financial sentiment annotations across thousands of financial documents.
- Constructed and validated a database of over 23,500 equity analysts by integrating Capital IQ records with publicly available professional biographies, institutional webpages, and CVs. Developed a Selenium/ChromeDriver entity-resolution pipeline that automatically investigated ambiguous analyst identities (e.g., abbreviated or alternate names), reducing 3,750 potential duplicate or uncertain matches to approximately 1,800 candidates requiring manual review.
- Performed manual verification of the remaining candidate matches, reducing the unresolved analyst count from 3,750 to just 54 records while substantially improving dataset completeness and entity-level matching accuracy for downstream empirical analyses.

Research Assistant, Dr. Tetsuji Yamada (Health Economics)

Jan 2023 – Dec 2025

Rutgers University

- Conducted a systematic review of more than 60 peer-reviewed studies to identify clinically relevant health economics measures, including quality-adjusted life years (QALYs), healthcare utilization metrics, and socioeconomic determinants, informing model specification for adolescent health behavior research.
- Developed generalized linear mixed models (GLMMs) with interaction effects to quantify associations between behavioral, demographic, and socioeconomic risk factors using Youth Risk Behavior Survey (YRBS) data, evaluating nested models through likelihood ratio tests, information criteria, and diagnostic analyses.
- Designed structural equation modeling (SEM) frameworks to investigate direct and indirect pathways underlying adolescent health outcomes, integrating latent behavioral constructs with observed demographic and socioeconomic variables to support causal hypothesis generation.

Research Assistant, Dr. Emara Noha (Labor Economics)

Jan 2023 – May 2023

Rutgers University

- Conducted a comprehensive review of the telecommunications and economic development literature to identify appropriate causal identification strategies, variable specifications, and empirical growth models, synthesizing findings to support the development of a research manuscript and conference presentation examining telecommunications infrastructure in Egypt.
- Implemented difference-in-differences (DiD) models in STATA to estimate the causal effects of telecommunications expansion on regional economic outcomes before and after major infrastructure investments, evaluating multiple treatment definitions, model specifications, and robustness checks based on methodologies reported in the applied econometrics literature.
- Developed instrumental variable (IV) estimation frameworks to address potential endogeneity between telecommunications investment and economic growth, assessing instrument validity through first-stage relevance tests, weak instrument diagnostics, overidentification tests, and specification comparisons to ensure reliable causal inference.
- Estimated the contribution of telecommunications infrastructure to long-run economic development using an augmented Solow growth model, systematically comparing alternative functional forms, interaction structures, and model specifications while interpreting their implications for productivity growth, labor market outcomes, and regional economic performance.

Teaching & Academic Experience

Teaching Assistant, Bayesian Statistics for Physical Sciences Johns Hopkins University

Jan 2026 – May 2026

- Designed, graded, and refined assignments covering Bayesian inference, probabilistic modeling, and decision theory, developing detailed solution keys and constructive feedback strategies to promote rigorous proof writing while improving grading consistency and fairness.
- Collaborated with the course instructor to design Bayesian exercises grounded in applications from the physical sciences, including particle detector experiments, astrophysics, and cosmology, helping connect statistical inference with realistic scientific problems.
- Produced lecture notes and computational demonstrations illustrating Bayesian computation, Approximate Bayesian Computation (ABC), and Markov chain Monte Carlo (MCMC) methods, while developing Mathematica simulations to visualize sampling algorithms and reinforce key computational concepts.

Teaching Assistant, Machine Learning Applications in Business Rutgers University

Aug 2024 – Dec 2024

- Supported instruction for two course sections totaling 74 students by designing and grading programming assignments covering supervised learning algorithms, predictive modeling, and model evaluation.
- Introduced GitHub Classroom workflows to automate coding assignment distribution and grading, demonstrating GitHub Classroom and Canvas integration to the course instructor to streamline assignment management and feedback.
- Established a student “DIY” technical support desk to troubleshoot GitHub, Anaconda, Zoom, and development environment issues, while manually editing Zoom lecture transcripts with timestamps, references, and formatting to improve accessibility and post-class review.
- Held weekly office hours providing individualized assistance with programming assignments, debugging, and machine learning concepts, while curating recent industry case studies and writing Canvas discussion posts highlighting real-world applications of machine learning in business analytics.

Statistics & Calculus Tutor Rutgers University

Jan 2022 – May 2025

- Delivered more than 150 tutoring sessions in statistics and calculus, translating mathematical concepts into accessible explanations while emphasizing their practical importance through discipline-specific examples in fields such as marketing, nursing, finance, economics, and the natural sciences.
- Designed and led supplemental instruction sessions focused on conceptual understanding, problem-solving strategies, and exam preparation, contributing to an approximately 15% increase in average exam scores among participating students.
- Mentored students on coursework, quantitative reasoning, and academic planning by adapting explanations to diverse learning styles and demonstrating how statistical and mathematical methods apply to real-world decision making across multiple disciplines.

Club Leadership Rutgers University

Oct 2022 – May 2025

- Advised organizational strategy and coordinated the planning and execution of more than 20 professional development, networking, and educational events, contributing to approximately 400% membership growth while mentoring new executive board members on event planning, leadership, and effective decision making under pressure.
- Maintained comprehensive organizational records, executive board schedules, meeting logs, and operational documentation while developing and continuously updating the organization’s internal handbook (“The Big Book of SFA”) to support leadership continuity, knowledge transfer, and long-term organizational sustainability.

- Designed and executed targeted outreach campaigns, optimizing recruitment materials, campus advertising, and communication strategies to maximize student engagement and strengthen relationships with faculty, alumni, and industry partners.

Organizations

Beta Alpha Psi, Beta Gamma Sigma, Omicron Delta Epsilon, Philadelphia Council of Business Economics

Projects

innovcal (Probabilistic Forecast Calibration under Misspecification)

GitHub

- Developed a probabilistic forecasting framework for studying forecast calibration under innovation misspecification by integrating score-based diffusion models with classical and deep multivariate forecasting architectures.
- Formulated diffusion-based innovation modeling by learning flexible residual distributions from forecast errors and recursively propagating sampled innovations through VAR and DeepAR systems to generate calibrated predictive distributions.
- Developed calibration-aware diffusion objectives by incorporating Continuous Ranked Probability Score (CRPS) directly into the training procedure and systematically evaluating differentiable Expected Calibration Error (ECE) surrogates proposed in the recent uncertainty quantification literature.
- Investigated calibration-aware modifications to the reverse diffusion process, including adaptive noise schedules, score-field regularization, residual dynamics, and timestep-dependent objective functions to explicitly improve probabilistic calibration rather than solely reconstruction accuracy.
- Constructed controlled stochastic simulation environments with Gaussian, heavy-tailed, skewed, mixture, and heteroskedastic innovation processes to isolate the effects of innovation misspecification on probabilistic forecast quality and robustness.
- Evaluated forecasting performance using PIT diagnostics, CRPS, Expected Calibration Error (ECE), energy scores, predictive interval coverage, interval width, and Wasserstein-based robustness analyses under controlled distributional shifts.
- Established both empirical and theoretical justification for calibration-aware diffusion forecasting through reproducible experimentation, ablation studies, and mathematical analysis of the proposed training objectives and reverse diffusion dynamics.

DiffIndEye (Diffusion-Based OCT Inpainting)

GitHub

- Formulated retinal OCT inpainting as a conditional inverse problem and developed a reproducible reconstruction framework spanning deterministic preprocessing, synthetic corruption, classical baselines, diffusion model training, and quantitative evaluation.
- Designed controlled horizontal-stripe masking experiments to generate anatomically meaningful missing-data problems with known ground truth, enabling rigorous comparison of reconstruction algorithms under standardized conditions.
- Implemented and benchmarked zero-fill, Gaussian smoothing, PDE-based diffusion, and Telea fast-marching inpainting against conditional denoising diffusion probabilistic models, analyzing reconstruction accuracy, convergence behavior, and computational tradeoffs across methods.
- Developed conditional encoder-decoder and U-Net DDPM architectures incorporating masked observations, diffusion timesteps, and corruption masks to investigate probabilistic anatomical reconstruction of retinal OCT images.
- Built a reproducible experimentation framework with deterministic preprocessing, checkpointing, experiment metadata, diffusion schedule validation, masked-region loss functions, and automated architecture benchmarking to support systematic numerical evaluation.
- Evaluated reconstruction quality using MSE, MAE, PSNR, masked-region diagnostics, retinal layer continuity,

and anatomical plausibility while comparing empirical performance against theoretical expectations reported in inverse problems and diffusion-model literature.

- Conducted numerical experiments to investigate theoretical limitations of diffusion-based reconstruction under structured missing-data regimes, demonstrating that optimization convergence alone was insufficient for anatomically faithful reconstruction; Telea inpainting achieved $\text{MSE} \approx 0.011$ and $\text{PSNR} \approx 28.6$, substantially outperforming both conditional DDPM architectures.

Live Opponent Adaptation in Reinforcement Learning

GitHub

- Developed a playable human-versus-agent Tron environment as a testbed for studying online reinforcement learning under nonstationary opponent behavior
- Implemented aggressive and defensive scripted agents for policy pretraining and trained a Proximal Policy Optimization (PPO) agent to establish baseline competitive behavior
- Designed session-level trajectory, action, reward, and outcome logging to preserve human gameplay experience for subsequent policy updates and longitudinal performance analysis
- Implemented checkpointing, model versioning, match replay, and session statistics to support reproducible evaluation of policy behavior across successive training stages
- Structured the online learning workflow so newly collected human gameplay can be incorporated into policy updates between sessions, enabling experiments on rapid opponent adaptation
- Designed the research framework to quantify adaptation speed and catastrophic forgetting while comparing replay-based continual learning, opponent-specific pretraining, and future robust or meta-reinforcement learning methods

SCART: Scientific Coding and Research Teammate

GitHub

- Developed a multi-agent AI research software assistant using LangGraph to support scientific computing, applied mathematics, statistics, machine learning, and research-oriented software development
- Designed a manager–worker architecture that classifies user requests and routes tasks across specialized agents for repository question answering, debugging, documentation, Git workflows, architecture review, code review, refactoring, dependency analysis, scientific workflow assessment, and web research
- Implemented a critic stage to evaluate and refine agent outputs for clarity, completeness, and consistency before returning responses to the user
- Built repository-level analysis workflows for identifying code smells, dependency inconsistencies, reproducibility issues, missing tests, architectural weaknesses, and maintainability concerns in scientific software projects
- Developed a lightweight evaluation framework with benchmark cases, routing validation, rubric-based response checks, keyword coverage diagnostics, and persistent evaluation reports
- Implemented observability infrastructure for structured logging, execution tracing, routing-frequency metrics, response statistics, and persistent experiment artifacts
- Designed the system as an extensible research platform supporting future AST-based repository intelligence, LLM-as-a-Judge evaluation, token and latency monitoring, additional model providers, and a standalone desktop interface

Personal Website AI Assistant | Demo

GitHub

- Developed a full-stack retrieval-augmented generation (RAG) system that answers questions about research, projects, coursework, and technical experience using semantically retrieved source documents
- Designed a modular AI assistant architecture with a Next.js/TypeScript frontend and FastAPI backend supporting document ingestion, chunking, embedding generation, semantic retrieval, prompt orchestration, and grounded LLM responses
- Implemented semantic search infrastructure using Supabase vector storage and metadata-aware retrieval to support scalable knowledge-base expansion and improved source selection

- Deployed the public-facing assistant using Vercel and Railway, incorporating API rate limiting, session controls, and responsive conversational UI/UX features
- Structured the system for future source-aware retrieval, metadata filtering, corpus scaling, evaluation workflows, and domain-specific research copilot extensions

Research Topic Lineages (Dynamic Topic Modeling + Diffusion)

GitHub

- Developed an end-to-end pipeline for modeling the temporal evolution of biomedical research topics using PubMed data, including automated ingestion, preprocessing, metadata management, and SQLite storage
- Generated transformer-based document embeddings and applied HDBSCAN clustering within individual time periods to discover evolving biomedical research themes without imposing a fixed topic structure
- Designed a temporal alignment framework that links related clusters across years into topic lineages, enabling measurement of topic persistence, emergence, disappearance, and structural change
- Quantified semantic drift through PCA trajectory analysis, cosine similarity, and movement norms to distinguish structural persistence from changes in topic representation over time
- Implemented a PyTorch diffusion model to learn stochastic dynamics over topic embedding trajectories and generate plausible future semantic trajectories under uncertainty
- Built Plotly and Matplotlib visualization workflows for lineage structure, movement distributions, temporal trajectories, and reduced-dimensional representations of high-dimensional topic evolution

Stock Forecast Dashboard

GitHub

- Developed an end-to-end neural time-series forecasting system using N-BEATS and N-HiTS architectures on financial data collected through the yfinance API
- Built a modular ingestion, preprocessing, and exploratory-analysis pipeline supporting missing-data handling, time-indexed feature construction, and technical indicator generation
- Implemented rolling-window and out-of-sample validation procedures to approximate real-world forecasting conditions while reducing lookahead bias
- Conducted systematic hyperparameter optimization across input windows, forecast horizons, and neural architecture configurations to improve predictive performance
- Benchmarked neural forecasting models against ARIMA and LSTM baselines, achieving an 18–25% reduction in RMSE across evaluated test periods
- Developed an interactive Plotly/Dash interface for exploring historical market data, model forecasts, and predictive outputs

Publications

Bayesian Hierarchical Modeling and Clustering for Malignant Cancer Diagnosis

2026

Argumentation Based Systems Journal

- Developed a 3-level Bayesian hierarchical logistic regression model (tumor, patient, region) to analyze late-stage cancer diagnosis using SEER data
- Performed full Bayesian inference using NUTS with convergence diagnostics (ESS , $\hat{R}$) and posterior predictive validation
- Quantified variance decomposition via intra-class correlations, showing dominant patient-level heterogeneity ($ICC = 0.94$) relative to region-level effects
- Applied Gaussian Mixture Models to posterior patient-level random effects, identifying latent risk clusters with distinct demographic and tumor profiles

Analyzing the Obesity of Children and Adolescents and the Effect of the Pandemic

2025

Celebration of Undergraduate Research and Creative Activity

- Analyzed adolescent obesity determinants using pooled NSCH data ($n \approx 39,700$) within the PRECEDE–PROCEED causal framework
- Estimated logistic regression models with average marginal effects to quantify impacts of socioeconomic status, healthcare access, and demographic factors
- Measured health inequality using Concentration and Kakwani indices, revealing disproportionate burden among lower-income and minority populations
- Built reproducible data pipelines in R/SQL, incorporating survey weighting, bootstrap inference, and diagnostic checks (VIF, reliability metrics)

Technical Skills

Programming: Python (PyTorch, TensorFlow, scikit-learn, sentence-transformers, statsmodels, lifelines, scikit-survival, MLflow, NumPy, pandas, polars, Numba, nltk, BeautifulSoup, SQLAlchemy, PostgreSQL, ArviZ), R (tidyverse, dplyr, lme4, survival, survminer, MatchIt, fixest, AER, ivreg, did, Synth, reticulate, BART, lavaan, ineq, CARET), SQL, Julia, C++, JavaScript, MATLAB, Mathematica

Statistical & Machine Learning Methods: Bayesian Computation, Nonparametric Bayesian Methods, Product Partition Models with Covariates (PPMx), Structural Equation Modeling, Generalized Linear Mixed Models, Survival Analysis, Multiple Time Series Analysis, Causal Inference, Instrumental Variables, Difference-in-Differences, Inverse Probability Weighting, Nonparametric Inference, Clustering Algorithms, ScoreGrad, TimeGrad, Diffusion Models, Reinforcement Learning

Optimization & Operations Research: Convex Optimization, Interior-Point Methods, Proximal Algorithms, Dynamic Programming, Integer Programming, Genetic Algorithms, Stochastic Optimization

Tools: STATA, SAS, LaTeX, Git, GitHub Classroom, Jupyter, Google Colab, ImageJ, Heidelberg Eye Explorer (HEYEX), Tableau, Weka, Jamovi, Bloomberg Terminal, Microsoft Office Suite

Certifications

CITI Good Clinical Practice (2026)

CITI Biomedical Research with Human Subjects (2026)

CFI Financial Modeling and Valuation Analyst (2024)

Bloomberg Market Concepts (2022)

Google Data Analytics (2023)

Columbia+ Machine Learning 1 (2025)

Research Interests

Robust and Reliable AI, Agentic Systems and LLM Reasoning, Bayesian Inference and Uncertainty Quantification, Probabilistic Forecasting and Calibration, Stochastic Generative Modeling, Reinforcement Learning and Sequential Decision-Making, Medical Image Analysis, Computational Biostatistics, Causal Inference